ARCHEGOS / CREDIT SUISSE

The March 2021 derivatives loss and the road to the March 2023 failure

1 Executive finding

Archegos was a major contributor, not the immediate trigger, of Credit Suisse's 2023 failure. Its March 2021 default caused approximately **$5.5bn of Credit Suisse losses** and exposed persistent weaknesses in client concentration controls, margining and escalation. The March 2023 failure was ultimately a **confidence-driven liquidity crisis** following several scandals and structural problems. [1, 2, 7]

What went wrong

Archegos used **total return swaps (TRS)** with several banks to build leveraged exposure to a small group of technology and media stocks. Banks hedged the swaps by holding underlying shares. As share prices fell in the week of 22 March 2021, Archegos could not meet collateral calls; dealers rushed to sell the hedges, deepening market declines. [3, 4, 5]

Event timeline

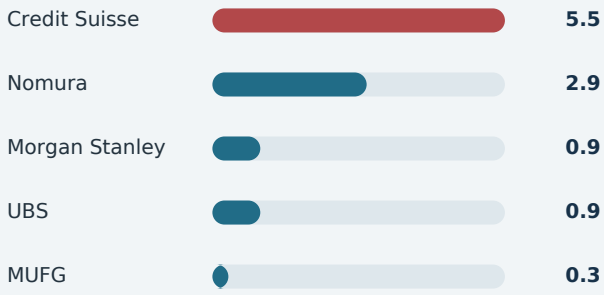
- **2019:** Credit Suisse cuts standard Archegos swap margin to 7.5%, down from previous 15–25% levels. [2]
- **2020:** Net-long concentration and repeated risk-limit overruns grow; exceptions and delayed remediation persist. [2, 6]
- **8 Mar 2021:** Risk committee identifies approximately $20bn swap book, 8–9% margin and single-name concentration. [2, 6]
- **23–26 Mar:** Underlying stocks fall; Archegos defaults on margin calls and banks liquidate hedges. [4, 5]
- **19 Mar 2023:** Swiss authorities arrange UBS acquisition amid acute confidence-driven liquidity outflows. [7, 8]

Scope: Credit Suisse group figures are distinguished from the $5.1bn booked at UK PRA-regulated entities. All derived ratios in this report are descriptive, not causal estimates.

2 Quantitative evidence

MEASURE	VALUE / IMPLICATION
Archegos position at CS, Mar 2021	**$24bn**; four times the next-largest hedge-fund client; >50% of CS Group equity. [1]
CS direct Archegos loss	**~$5.5bn** group-wide; $5.1bn in PRA-regulated UK firms. [2, 6]
2020 revenue from Archegos	**$17.4m**, based on CS committee materials. [2]
Risk exposure, 23 Feb 2021	**$610m** potential exposure (PE) vs recently raised $50m PE limit. [6]
Early Mar swap margin	**8–9%**; February dynamic proposal would yield 16.74%. [6]
Margin released, 11–19 Mar	**$2.4bn** variation margin remitted under existing contracts. [2, 6]

Reported bank losses (USD bn)



ESMA Chart 1; disclosed loss estimates as of April 2021 [3]

Derived scale indicators: CS accounts for **52.4%** of the five plotted banks' $10.5bn disclosed losses (5.5/10.5); the $5.5bn loss was **~316 times** Archegos-related CS revenue reported for 2020 (5,500/17.4). The latter is a cross-period scale comparison, not an investment-return measure. [2, 3]

3 How the controls failed

A. Static initial margin under-captured the risk

For many swaps, the dollar amount of initial margin was fixed when the trade began. As the market value of the positions rose, effective margin coverage declined. The U.S. FSOC illustrates how $20 collateral on a position rising from $100 to $140 falls from **20% to 14.3%** coverage; the same mechanism eroded Archegos's protection. The FSOC reported an average Credit Suisse swap margin of **6.9% in December 2020**. [5]

B. Client terms overrode risk limits

Despite repeated breaches, risk limits were relaxed rather than positions reduced. Archegos's internal rating was downgraded from BB- to B+, but its **PE limit was raised from $20m to $50m**. On 23 February 2021, its measured PE was $610m: **12.2x the revised limit**. This compares an internal risk metric to an internal tolerance, not expected realized loss. [6]

C. Dynamic margining came too late

Credit Suisse calculated that applying its existing Prime Brokerage-style dynamic rules would require **$3.0bn additional initial margin**. Its less stringent February proposal would have required about **$1.27bn**, yet the March 8 committee contemplated only a **$250m** request if negotiations stalled: roughly **8.3%** of the stricter $3bn shortfall. Dynamic margining was not implemented before default. [2, 6]

D. Weak escalation and governance

FINMA reported that group executive-board members were not adequately informed of the exceptionally large client position. Credit Suisse repeatedly raised limits or allowed overruns and lacked appropriate concentration-aware collateral. The PRA also found no evidence that Credit Suisse considered using its right to increase initial margin with three days' notice before paying $2.4bn variation margin. [1, 6]

Important nuance: the PRA records that the $2.4bn payments were required by the swaps contract. Their release increased vulnerability, but should not be described as an unlawful or wholly avoidable transfer. [6]

4 Exposure and default mechanism

Hidden leverage and issuer concentration

ESMA's EU trade-repository study observes **~6x synthetic leverage**. Its partial EEA sample shows that on 26 February 2021 the top five long stocks made up **80% of long exposures**, while gross exposure to EU counterparties was **2.5x** end-2020 levels and net exposure **7x**. The analysis does not cover all global counterparties; Brexit-related reporting breaks also affect the plotted series. [3]

Mechanics of the loss

- **1.** Concentrated stock prices fall; TRS positions move against Archegos.
- **2.** Banks issue variation-margin calls; Archegos cannot fund them (CFTC alleged aggregate calls exceeding $13bn).
- **3.** The default forces liquidation of the banks' physical equity hedges.
- **4.** Simultaneous block sales deepen the price shock, creating collateral shortfalls and realized credit losses.

Fraud and institutional responsibility

The **CFTC 2022 complaint alleges** that Archegos personnel misled counterparties about gross exposure, cash, liquidity and concentration, allowing additional financing on favorable terms. These are allegations in a pleading, not standalone proof that every charge was adjudicated. Independently, the 2021 Credit Suisse board review and UK/Swiss supervisory findings document preventable deficiencies inside the bank, including overlooked warnings and persistently ineffective margin controls. [1, 2, 4, 6]

Was Archegos systemically destabilizing in 2021?

No according to ESMA and the U.S. FSOC: in the calmer 2021 markets, individual major banks absorbed the exposures without systemic contagion. This reinforces the distinction between a **severe bank-specific loss in 2021** and the later **systemic emergency in 2023**. [3, 5]

5 Data-analytics investigation design

Objective: detect non-disclosed leverage, risk-limit accommodation, under-collateralization and anomalous exposure growth before liquidation risk becomes acute. Available public documents support a retrospective risk design but do not contain the trade-level observations needed to estimate model precision or backtest specific early alerts.

Requested datasets

• Daily TRS notionals, underlying identifiers, collateral, margin-rate schedules and hedge inventory by legal entity / prime broker.

• Potential-exposure and stress-scenario reports; limits, exceptions, approvals, and escalation timestamps.

• Client NAV and gross/net leverage, independent counterparty attestations, trade-repository reports and cash movements.

• Orders, forced-sale market volumes, execution timestamps, liquidity scores and internal communications.

Detection logic (proposed, not executed)

• **Rule flags:** PE/limit >1; repeat breaches >30 days; concentration in top five names; static margin ratio falling while exposure rises.

• **Time series:** rolling 20-day exposure growth, collateral shortfall vs dynamic margin, jump in effective leverage, liquidity-adjusted days to exit.

• **Entity graph:** shared issuers and mirrored positions across brokers; link each risk exception to decision owners.

• **Stress replay:** synchronized multi-bank margin calls and price-impact liquidation simulation on concentrated names.

Validation and limitations

Reconstruct daily portfolios and check trade breaks against custody, finance and repository snapshots. Perform rolling-origin backtests and evaluate **precision, recall, false-positive burden, lead time, and stressed loss coverage**. Compare model alerts with documented events in September 2020, February 2021 and March 2021. Main limitations are confidential broker data, non-EEA reporting gaps, rapidly changing collateral terms, and hindsight bias. [3, 6]

6 Link to the March 2023 failure

Measured direct impact (2021)

The **$5.5bn loss** directly reduced Credit Suisse's earnings and capital-generating capacity, prompting substantial remediation and regulatory scrutiny. The Archegos default also temporarily put Credit Suisse International below its combined liquidity coverage threshold: **95.5%** at March 2021 month-end versus the **100%** requirement, according to the PRA. That breach was corrected shortly afterward. [2, 6]

Indirect deterioration (2021–2023)

Archegos followed the Greensill fund crisis and added to a sequence of scandals, business losses and management failures that undermined the franchise. FINMA attributes the collapse to cumulative strategic and governance failures that ultimately drove withdrawals. In early October 2022 the FSB records client/counterparty outflows as high as **CHF 12–15bn per day**; renewed outflows after U.S. bank failures in March 2023 led to an emergency UBS merger announced **19 March 2023**. The Archegos episode was one contributing factor, **not a sufficient or sole cause**. [7, 8]

Key conclusions

• **Excessive concentration:** an unusually large single-client book made a hedge strategy vulnerable to coordinated liquidation.

• **Ignored early warnings:** known breaches and low margins were not translated into effective restrictions.

• **Transparency failure:** multiple dealers lacked a complete view of a highly leveraged family office.

• **Cumulative franchise damage:** major direct losses and management failures compounded a later crisis of confidence.

Selected source notes (numbers refer to the files supplied)

[1] FINMA, Archegos: FINMA concludes proceedings (24 Jul 2023), pp. 1–3.

[2] Credit Suisse Special Committee, Report on Archegos (29 Jul 2021), pp. 1–23, 117–120 (report pagination).

[3] ESMA, Leverage and derivatives – the case of Archegos (May 2022), pp. 3–9.

[4] CFTC, Complaint against Archegos and Halligan (27 Apr 2022), pp. 1–4.

[5] FSOC, 2021 Annual Report, Box D, pp. 109–110.

[6] PRA, Final Notice to CSI and CSSEL (21 Jul 2023), §§2, 14–16.

[7] FINMA, Lessons learned from the Credit Suisse crisis (19 Dec 2023), pp. 1–2.

[8] FSB, 2023 Bank Failures (10 Oct 2023), pp. 5–7.